Motivation Letter: Research Engineer - Language Models for Science (RE2)

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Recruitment Panel
Barcelona Supercomputing Center - Centro Nacional de Supercomputación

Subject: Application for Research Engineer - Language Models for Science (RE2), reference 397_26_AII_S_RE2

Dear Members of the Recruitment Panel,

I am applying for the Research Engineer position in Language Models for Science because the EVELYN project's combination of foundation models, heterogeneous scientific data, and environmental applications fits my interdisciplinary training. I recently completed an MSc in Modelling for Science and Engineering at the Universitat Autonoma de Barcelona, specialising in Data Science, after a BSc in Mechanical Engineering. I completed my BSc under the Hungarian Government's Stipendium Hungaricum scholarship, which covered my tuition and provided living and accommodation support. I was later selected for CSIC's JAE Intro ICU 2025 research-introduction scholarship at IIIA-CSIC, where I developed my MSc thesis. My work has progressed from machine learning for a physical system to enterprise language-model applications and a complete agentic robotics thesis.

At Aily Labs, I took responsibility for integration seams connecting go-to-market data capabilities to SuperAgent, the company's general-purpose, customer-facing LLM reasoning agent. My work progressed from MCP server tools to task-scoped skills and then into a governed semantic layer. Curated Markdown resources encoded the business meaning, relationships, permitted use, and access constraints of approved SQL tables so specialised agents could select the right data without relying on raw schemas. I developed and maintained GTM capabilities used by Aily tenants including Sanofi.

When answer quality regressed or a skill, resource mapping, or workflow contract changed, my team was responsible for evaluating and correcting the behaviour from development through production. My contribution was to trace failures across data availability, semantic descriptions, table allowlists, tool selection, DAG payloads, and post-processing, then verify the correction before promotion. Code review, automated testing, representative question sets, and trace-based checks gave me practical experience with production language-model behaviour, metadata quality, non-deterministic evaluation, and reliable model-assisted services.

My machine-learning foundation began with physical modelling. My BSc thesis built and validated a neural network for turbulent-intensity prediction from wind-profile data, requiring data preparation, model design, validation, and interpretation against the underlying fluid-mechanical problem. At ABS Consulting, I later supported seismic qualification and structural assessment for UK nuclear power assets. That work reinforced reviewable computation, standards-based reasoning, and precise technical reporting in a domain where results must remain physically and procedurally defensible.

The Data Science specialisation strengthened this background through large-data modelling and distributed computation. Parallel Programming covered C, CUDA, OpenMP, MPI, SLURM, GPU programming, and performance evaluation. Distributed Systems covered cluster and supercomputer infrastructure, heterogeneous batch workloads, cloud systems, distributed databases, and fault tolerance. I used these methods in laboratory and project work to submit and monitor cluster jobs, develop parallel solutions, and assess computational behaviour.

iTrader is my autonomous trading platform under staged development. Its implemented offline stack learns from OHLCV market windows and solver-specific engineered features rather than hand-authored trading labels. It combines validated task specifications, PPO/GAE, task-conditioned FiLM policy and value networks, finite decomposed rewards, local language-model proposers, and reproducible run manifests. The online design progresses through replay and shadow operation before live execution, with deterministic decision and risk gates. The project has strengthened my PyTorch practice, reward design, experiment control, and ability to detect exploitation of an imperfect objective.

My MSc thesis at IIIA-CSIC focused on a modular language-model-enabled interaction and execution stack for NAO. I integrated dialogue, grounded symbolic knowledge, structured planning, deterministic action dispatch, execution feedback, and replanning through ROS 2 and ROS4HRI. The scenario-based validation framework used structured traces, knowledge-base postconditions, and controlled failure policies. Although the application is robotics rather than environmental science, it required heterogeneous information to be transformed into bounded model context and evaluated against observable outcomes.

Watson adds practical model-serving experience through a personal agent platform that I operate for development and research. Hermes provides the agent harness and tool workflow, LiteLLM provides routing and an OpenAI-compatible proxy, llama.cpp serves a quantised Qwen model on the GPU, and ZeroTier exposes the endpoint across my machines. I maintain its launch, health, recovery, and routing paths and measure context size, time to first token, throughput, memory use, and tool-call fidelity. This complements my professional work by exposing model-serving trade-offs and the need for reproducible interfaces between models and domain data.

My direct experience is currently stronger in agentic language-model systems, reinforcement learning, and scientific modelling than in training large multimodal foundation models end to end. I would bring the underlying skills needed to grow into that responsibility: Python and Linux development, PyTorch-based model work, structured data processing, evaluation design, scientific reading, and computational training that includes GPU and cluster workflows. EVELYN is particularly attractive because it applies these methods to environmental pollution, where heterogeneous observations and physical interpretation both matter.

I work independently, document decisions carefully, and value direct technical discussion in multidisciplinary teams. I am fluent in English and Spanish and have studied or worked in Hungary, the United Kingdom, and Spain. I would welcome the opportunity to contribute to model development, data processing, experimentation, and scientific communication within BSC's AI Institute.

Thank you for considering my application. I would be pleased to discuss how my language-model engineering, scientific ML background, and computational training could contribute to EVELYN.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Francisco Martin**, Principal Data Scientist, Aily Labs

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2. **Dr. Balogh Miklos**, Professor, BME

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